



Indian Association for the Cultivation of Science
(Deemed to be University under the *de novo* category)

BS-MS Program

End-Semester Examination-2025 (Autumn Semester-I)

Subject: Probability I

Subject Code(s): MAT 2101

Full marks: 50

Time allotted: 3 hrs

Answer question number 10 and any 6 of the remaining questions.

1. Suppose X is a bounded random variable, i.e., there exists a constant k such that $P(|X| \leq k) = 1$, then show that X has moments of every order. [7]

2. The joint pdf of the random variables X, Y, Z is given by

$$f(x, y, z) = 8xyz, \quad 0 < x, y, z < 1.$$

Find $P(X < Y < Z)$. [7]

3. Let X be a continuous random variable with pdf $f > 0$. Show that $E|X - c|$ becomes minimum when $c =$ the median of X . [7]

4. Let X be a continuous non-negative random variable with finite mean. Show that

$$EX = \int_0^\infty [1 - F(x)] dx$$

where F is the cdf of X . [7]

5. Let $X_1, \dots, X_n$ be independent, identically distributed random variables with finite expectation, and $Y = X_1 + \dots + X_n$. Show that

$$E[X_1|Y] = \frac{Y}{n}.$$

[7]

6. Let $X_1, \dots, X_n$ be independent random variables. Suppose that X_i has Poisson distribution with parameter λ_i , $i = 1, \dots, n$. Show that $X_1 + \dots + X_n$ is Poisson distributed with parameter $\lambda_1 + \dots + \lambda_n$. [7]

7. Let $X_1, X_2, \dots$ be a sequence of iid exponential random variables with parameter λ , and N a geometric random variable with parameter p , independent of $X_1, X_2, \dots$. Let $Y = X_1 + X_2 + \dots + X_N$. Find the pdf, expectation and variance of Y . [7]
8. Let X and Y be dependent random variables with common mean 0 and variance 1. Let ρ be the correlation coefficient. Show that

$$E[\max(X^2, Y^2)] \leq 1 + \sqrt{1 - \rho^2}.$$

[7]

9. Show that

$$\lim_{n \rightarrow \infty} e^{-n} \sum_{k=0}^n \frac{n^k}{k!} = \frac{1}{2}.$$

[7]

10. Let X be uniformly distributed in the interval $[4, 10]$ and suppose that we observe X with some random additive error W , i.e., we observe the value of the random variable

$$Y = X + W.$$

We assume that W is uniformly distributed in the interval $[-1, 1]$, and independent of X . What is the linear least square estimate of X given Y . [8]